Project Proposal: Bayesian Estimation

Stella Rovazzi · Ramez Elprince · Brenden Doogan

University of Guelph

Project Overview

Objectives

- bleh [3]

0.1 Background and Literature

1 Proposed Computational Strategy

Simple Example

Goal

We will start with a simple example to ensure we can correctly converge to some posterior. Alongside this, we will ensure we can calculate (and infer) posterior summary statistics, suitability of the model, monte-carlo standard error, and rate of convergence.

Methodology

This follows the example C.3 regarding the normals in Bayesain Data Analysis (BDA) [1].

1. Generate bivariate normal data
2. Derive and make functions for the marginal distributions (assuming normal prior and conditional)
3. Run the posterior simulations with Gibbs Sampling
4. Find the parameters through the samples
5. Repeat for various realistic n

Harder Example

Goal

Apply what was learned in the simple example to real data. We will be able to try and compare the our methodology.

Methodology

This follows the bioessay expirement in example 3.7 from BDA [1].

1. Derive the full conditional, then the marginal conditionals using the distributions provided in text
2. Gibbs sample for the posterior
3. Explore the results
4. Repeat for various realistic n

Extend to Hierarchical Models

If time permits, we would use the rat tumor example (5.4) from BDA and tutorial C.3 to try to construct a bayesian hierarchical model using stan [1].

Goal

This would extend our knowledge to learn hyperparameters and Stan.

1. Create stan file, laying out parameters, model, and transformed parameters
2. Run some number of chains iteratively in R, which can be set to be the Gibbs method
3. Use Stan and R to find the posterior

Quantities of Interest

- Estimated mean, variance/standard deviation, credible intervals (informs us of the posterior model) [1]
- Gelman-Rubin Statistic (Convergence Assessment) [4]
- Rate of Convergence (/Assessing with Gibbs Stopper) [5]

Required Libraries

- MASS for the bivariate normal generation (or basic R) [3]
- STAN for the actual MCMC/Gibs sampling – we may some examples by hand [2]

Division of Responsibility

Timeline

We can chat about GANTT charts on tuesday or smth, just so we can layout realistic dates with buffer room.

Expected Outcomes

bleh

References

[1] Andrew Gelman, John B. Carlin, Hal S. Stern, David B. Dunson, Aki Vehtari, and Donald B. Rubin. *Bayesian data analysis*. Texts in statistical science. CRC Press, Taylor & Francis Group, Boca Raton, third edition edition, 2014. ISBN 978-1-4398-4095-5 978-1-4398-9820-8.

[2] Jiqiang Guo, Jonah Gabry, Ben Goodrich, Andrew Johnson, Sebastian Weber, and Hamada S. Badr. rstan: R Interface to Stan, July 2015. URL `https://CRAN.R-project.org/package=rstan`. Institution: Comprehensive R Archive Network Pages: 2.32.7.

[3] Brian Ripley and Bill Venables. MASS: Support Functions and Datasets for Venables and Ripley's MASS, May 2009. URL `https://CRAN.R-project.org/package=MASS`. Institution: Comprehensive R Archive Network Pages: 7.3-66.

[4] Maria L. Rizzo. *Statistical computing with R*. CRC Press, Taylor & Francis Group, Boca Raton, second edition edition, 2019. ISBN 978-1-4665-5332-3 978-1-4665-5333-0.

[5] Martin A. Tanner. *Tools for Statistical Inference: Methods for the Exploration of Posterior Distributions and Likelihood Functions*. Springer Series in Statistics. Springer New York, New York, NY, 1996. ISBN 978-1-4612-8471-0 978-1-4612-4024-2. doi: 10.1007/978-1-4612-4024-2. URL `http://link.springer.com/10.1007/978-1-4612-4024-2`.